# Zi En Tham (Zen)

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## Education

**Bachelor of Computing Science (Honours)** **August 2021 - August 2024**
**University of Technology Sydney (UTS)**

- Major: Artificial Intelligence and Data Analytics (WAM 86.58)

## Skills

**Programming language:** SQL, Python, HTML/CSS, Java, PHP, Javascript, C++
**Languages:** English, Chinese (Native), Cantonese, Malay
**Tools / Tools:** Git, Node, React, TypeScript, TailWindCSS, Yi Framework, Laravel, Excel, Tableau, PowerBI

## Work Experience

**Web Developer** **May 2025 – Present**
**Trivesta, Sydney, NSW, Australia**

- Built and maintained full-stack features for a corporate website and internal CRM.
- Built n8n automation workflows to handle client calls and inquiries, reducing manual processing time.
- Built predictive AI/ML churn models achieving 80%+ accuracy, and directly informing client retention strategies.
- Tools: Wordpress, Typescript, React, Vite, PHP, Javascript, Redux, Axios, MySQL, AntDesign, TailwindCSS, n8n, PowerBI

**Operation Developer** **September 2024 - Present**
**Flash Media Solutions, Rydalmere, NSW, Australia**

- Developed a modern e-commerce website serving over 20 products and 50 monthly visitors, migrating from a legacy stack to improve performance and maintainability
- Built an automated Excel stock and invoice tracking system that eliminated duplicate records across 250 inventory items.
- Designed Tableau dashboards visualizing sales trends, regional client distribution, and stock velocity.
- Managed client relationships and translated business requirements into technical solutions.
- Tools: PHP, JavaScript, GSAP, WordPress, Excel, Tableau

**Research Assistant** **August 2024 - December 2024**
**La Trobe University, Melbourne, Victoria, Australia**

- Developed co-learning MedNext based model for brain haemorrhage detection, achieving 86% accuracy in classifying brain hemorrhage in CT scans.
- Adapted Bal-CE loss and post-hoc logit adjustment to improve class balance and achieved up to 21% improvement in accuracy and 20% in ROC AUC over previous state-of-the-art methods.
- Tools: Python, Docker, Scikit-learn, MedNext, CT Imaging

**U: PASS Data Team** **January 2023 - June 2024**
**University of Technology Sydney, Sydney, NSW, Australia**

- Collected, cleaned, and maintained attendance records for over 200 sessions, ensuring data integrity for academic performance reporting.
- Generated Excel reports and visualizations tracking study pattern trends and grade correlations, used by supervisor for program evaluation.
- Tools: Microsoft Excel

## Projects

**Anatomy of Kickstarter** **February 2026**

- Analysed 200,000 Kickstarter campaigns spanning a full decade (2009–2020)
- Filtered advertising-disguised campaigns to surface genuine fundraising patterns and trends
- Visualized platform genre evolution from creative category dominance to tech-games parity
- Tools: Tableau, Data Analysis, Data Cleaning

**Inventory Management System** **October 2025**

- Designed normalized Excel VBA database to replace a broken legacy CSV system
- Automated stock validation, warranty tracking, and aging inventory alert generation
- Eliminated duplicate records across 30 product SKUs entirely
- Tools: Excel VBA, Database Design, CSV Processing

**Australia Homelessness Analysis** **April 2025**

- Analysed multi-source datasets revealing Victoria's disproportionately high homelessness rate
- Identified mental health as the primary driver among the 35–44 age group
- Published interactive Tableau dashboard enabling public exploration of key findings
- Tools: Tableau, Data Analysis, Public Data

**Australia Economic Trade Analysis** **September 2023**

- Analyzed 34 years of Australian international trade data using Tableau visualisations
- Focused on machinery and crude materials to surface meaningful long-term trade trends
- Delivered stakeholder-ready insights report for economic and policy decision-makers
- Tools: Tableau, Trade Data Analysis, Economic Research

**Property Analysis in Mortdale** **August 2023**

- Cleaned and analyzed ABS housing data to examine Mortdale suburb price trends
- Correlated property values with key demographic and community characteristic variables
- Produced actionable buyer and renter recommendations report for academic submission
- Tools: Excel, Data Analysis, ABS Data

**Urban Heat Index Prediction for Clinical Laboratory (EY Open Science 2025)** **March 2025**

- Built a Random Forest model predicting Urban Heat Index in Manhattan and Bronx
- Achieved 97% $R^2$ score through manual sourcing and integration of diverse datasets
- Extracted, mapped, and processed geospatial, time series, and numerical data formats
- Tools: Python, Random Forest, GeoPandas, Time Series Analysis

**Anomaly Detection with LSTM for Clinical Laboratory (Honours Thesis)** **June 2024**

- Built unsupervised LSTM system for real-time anomaly detection in clinical lab data
- Combined statistical QC methods with deep learning to enable proactive error detection
- Achieved 5-10% improvement over traditional QC baselines
- Tools: Python, LSTM, Neural Networks, Statistical Methods

## Referees

Dr Xu Wang
Senior Lecturer at School of Electrical and Data Engineering UTS
[Contact details available on request]

Eugene Nguyen
Formal Colleague at Trivesta
[Contact details available on request]